

2. Requirement Analysis Phase

- The Requirement Analysis Phase identifies the functional and technical requirements necessary to implement the spreadsheet data-import project. At this stage, the requirements are studied carefully before actual development begins. Proper requirement analysis helps ensure that the final import process satisfies the expected business and technical objectives.
- The first requirement is to identify the source data. In this project, the source is a spreadsheet, generally in Excel or CSV format. The spreadsheet should contain properly organized columns and rows. Each column should represent a particular field, while each row should represent one record. For example, an employee spreadsheet could contain Employee ID, Name, Email, Department, Location, and Joining Date.
- The next requirement is to identify the target table. The target table is the location where the imported records will finally be stored. The target table must contain appropriate fields corresponding to the source spreadsheet columns. If the source and target field names are different, the Transform Map must establish the relationship between them.
- Another important requirement is the Import Set or equivalent staging mechanism. Imported spreadsheet data should first be placed into a temporary or staging table before being transformed into the target table. This provides an intermediate layer where the source data can be reviewed and validated.
- The project also requires a Transform Map. The Transform Map defines how data from the source table is transferred to the target table. It can contain field mappings, coalesce fields, transformation scripts, and other rules. A coalesce field can be used to determine whether an incoming record should create a new record or update an existing record.
- Data validation is another major requirement. The project should verify whether mandatory fields are present, whether email addresses have valid formats, whether dates are correctly formatted, and whether duplicate records exist. Invalid data should be identified before or during the transformation process.
- Security and access requirements should also be considered. Only authorized users should be able to upload spreadsheet data or modify Transform Maps. The user performing the import should have the required permissions to access the import functionality, staging table, target table, and transformation configuration.
- The major requirements can therefore be summarized as:
 - A properly formatted spreadsheet.
 - A suitable source/import table.

A suitable target table.

A configured Transform Map.

Correct source-to-target field mappings.

Data validation rules.

Duplicate-record handling.

Transformation logic where required.

Testing and error handling.

Proper documentation of the import process.

Requirement analysis provides a clear understanding of what must be developed and tested. It also reduces the possibility of errors during implementation.